

Erica F. Huang Wellesley, MA | (781) 985-0004 | erica.f.huang@gmail.com | <https://github.com/erica-f-huang>

EDUCATION

Tufts University, Medford, MA

May 2027

B.S.E. in Computer Science, Minor in Embedded Systems

3.88/4.0 GPA, Dean's List (5 semesters)

Relevant Courses: Data Structures, AI: Algorithms, Ethics, & Policy, Machine Structure & Assembly Language Programming, Embedded Systems, Human Computer Interaction, Digital Circuits, Technical Communication, Security

EXPERIENCE

Research Assistant (Paid), Tufts University Human Robot Interaction Lab, Medford, MA

May 2025 – Present

- Training Boston Dynamics' Spot robot to improve object manipulation skills in an open-world environment
- Running experiments to understand how teams work to create an LLM to assist teamwork

Software Engineering Intern (Paid), Cytel, Wellesley, MA

Summer 2024, Summer 2025

- Optimized code using parallel processing and communicating between R and C++ for faster runtime
- Generated reports using simulation output in R to provide clients with a formatted summary of their simulated trials
- Tested different forms of optimization (using a dynamically linked library to communicate between C++ and R, running C++ in parallel, and running R in parallel) to understand which method results in the fastest runtime
- Presented well-received analysis and recommendations to a team of senior developers for further development of my work

Founder and Lead Student Mentor, FIRST LEGO League (FLL) Team, Wellesley, MA

May 2022 – May 2023

- Created and led a partnership with Wellesley Middle School science and technology teachers to launch two FIRST LEGO League Challenge teams for 6th through 8th-grade students
- Placed 9th and 19th out of 36 teams at the 2022 regional competition
- Won the team award for displaying FIRST core values
- Currently a team advisor and mentor during university breaks for around 20 students

Founding Member and Coding Lead, FIRST Robotics Competition (FRC), Wellesley, MA

Aug 2020 – May 2022

- Led a nine-member coding team, contributing to an award-winning robot
- Expanded and gender-diversified coding team, contributing to a second award-winning robot
- Established and served in the team's first Director of Community Education and Outreach role
- Created and implemented summer robotics workshops for 3rd and 4th graders and collaborated with local Girl Scout leaders to develop and run badge-earning events

AWARDS AND ACHIEVEMENTS

Centennial Youth Service Award

2023

Awarded by the town of Wellesley for founding and leading two FIRST LEGO League teams for the town's public middle school.

Dean's List Nominee, FIRST Robotics Competition

2022

Nominated for leadership and professionalism by my 40-member team and mentors from the community and industry.

SKILLS, INTERESTS, HOBBIES, EXTRACURRICULARS

Skills: C++, C, Java, R, Python, LaTeX, robotics, human-robot interaction, parallel computing, optimization, neural networks, digital circuits, digital electronics, embedded systems, human-computer interaction, Linux, Git

Interests: Robotics, human-robot interaction, artificial intelligence, programming, machine learning, mechanical and electrical systems design, physics, calculus, performance optimization, ethical applications of technology, embedded systems

Hobbies: Snowboarding, weightlifting, basketball, triathlons, rock climbing, backpacking, puzzles, saxophone

Tufts Clubs: Basketball, Wilderness Orientation, Mountain Club, Dance Collective, Outreach Learning Fellows, Robotics